

Data Centers Project

Scalable Architecture for Multi-Departmental Campus

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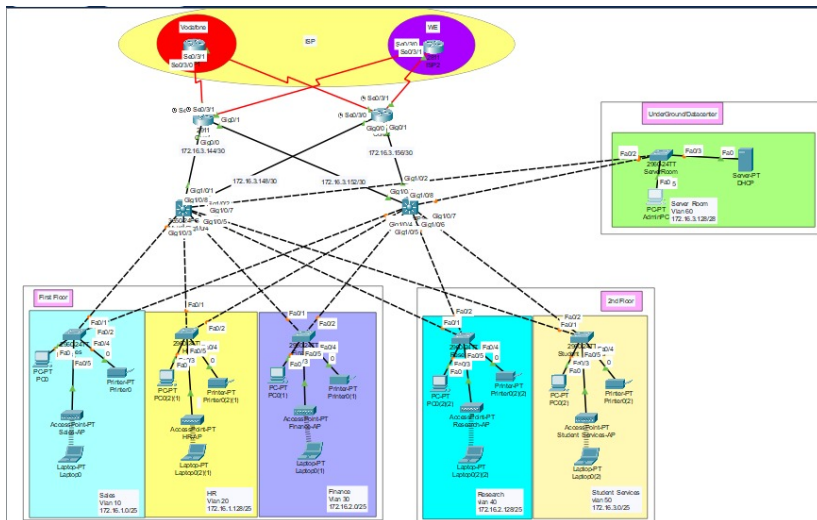
Course: IT 401 - Data Centers

Objective: Design a robust network infrastructure for a 3-story building supporting 600+ users.

- **Departments:** Sales, HR, Finance, Research, Student Services.
- **Critical Assets:** Dedicated Server Room (Underground).
- **Key Requirements:**
 - High Availability (Dual ISP Redundancy).
 - Logical Segmentation (VLANs).
 - Dynamic Routing (OSPF).
 - Centralized Management (SNMP).

Network Topology Design

The design utilizes a hierarchical 3-tier architecture (Core, Distribution, Access) to ensure scalability.



IP Addressing Strategy

We utilized **VLSM** to efficiently allocate the 172.16.1.0/22 block.

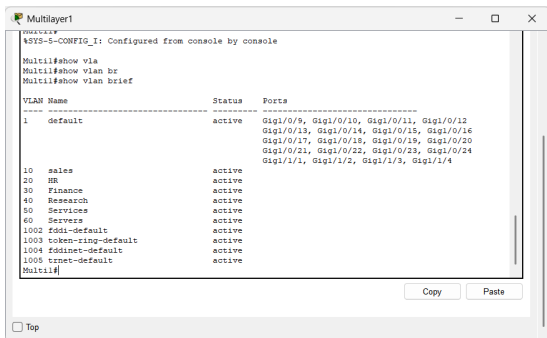
Dept	VLAN	Subnet	Mask
Sales	10	172.16.1.0	/25
HR	20	172.16.1.128	/25
Finance	30	172.16.2.0	/25
Research	40	172.16.2.128	/25
Student Svcs	50	172.16.3.0	/25
Server Room	60	172.16.3.128	/28

Table: VLAN & IP Allocation Table

Implementation: VLAN Segmentation

Security through Isolation: Each department resides in a separate broadcast domain.

- VLANs 10-60 Created.
- Inter-VLAN routing via Multilayer Switches (SVIs).



```

Multilayer1
%SYS-5-CONFIG_I: Configured from console by console

Multil#show vla
Multil#show vlan br
Multil#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
                               Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
                               Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
                               Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
                               Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   sales                    active
20   HR                       active
30   Finance                  active
40   Research                 active
50   Services                 active
60   Servers                  active
1002 fddi-default            active
1003 token-ring-default      active
1004 fddinet-default         active
1005 trnet-default           active
Multil#

```

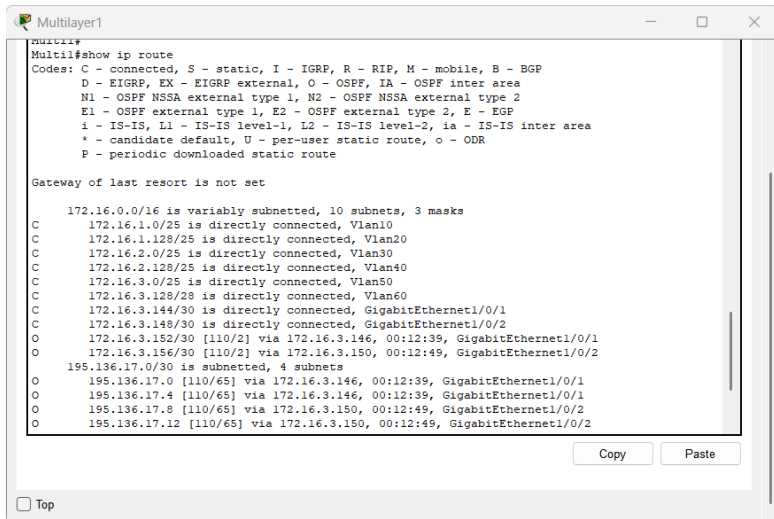
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Figure: VLAN Database Verification

Implementation: OSPF Routing

Dynamic Redundancy: OSPF Area 0 configured on Core Routers and Distribution Switches.



```
Multilayer1
Multil#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 10 subnets, 3 masks
C       172.16.1.0/25 is directly connected, Vlan10
C       172.16.1.128/25 is directly connected, Vlan20
C       172.16.2.0/25 is directly connected, Vlan30
C       172.16.2.128/25 is directly connected, Vlan40
C       172.16.3.0/25 is directly connected, Vlan50
C       172.16.3.128/28 is directly connected, Vlan60
C       172.16.3.144/30 is directly connected, GigabitEthernet1/0/1
C       172.16.3.148/30 is directly connected, GigabitEthernet1/0/2
O       172.16.3.152/30 [110/2] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       172.16.3.156/30 [110/2] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
195.136.17.0/30 is subnetted, 4 subnets
O       195.136.17.0 [110/65] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       195.136.17.4 [110/65] via 172.16.3.146, 00:12:39, GigabitEthernet1/0/1
O       195.136.17.8 [110/65] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
O       195.136.17.12 [110/65] via 172.16.3.150, 00:12:49, GigabitEthernet1/0/2
```

Services: DHCP Configuration

Automated Addressing: Centralized DHCP Server located in VLAN 60 (Server Room).

Physical Config **SERVICES** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

Start IP Address: 172.16.2.129

Subnet Mask: 255.255.255.240

Maximum Number of Users: 512

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
ResearchPool	172.16.2.129	172.16.3.131	172.16.2.134	255.255.255.128	120	0.0.0.0	0.0.0.0
FinancePool	172.16.2.1	172.16.3.131	172.16.2.6	255.255.255.128	120	0.0.0.0	0.0.0.0
Student_ServicesPool	172.16.3.1	172.16.3.131	172.16.3.6	255.255.255.128	120	0.0.0.0	0.0.0.0
HRPool	172.16.1.129	172.16.3.131	172.16.1.134	255.255.255.128	120	0.0.0.0	0.0.0.0
SalesPool	172.16.1.1	172.16.3.131	172.16.1.6	255.255.255.128	120	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	172.16.3.128	255.255.255.240	512	0.0.0.0	0.0.0.0

Management: SNMP Implementation

Proactive Monitoring: SNMPv2c deployed for remote device monitoring.

Configuration:

- Community: public (RO)
- Manager: Admin PC
- Target: Core/Distribution Devices

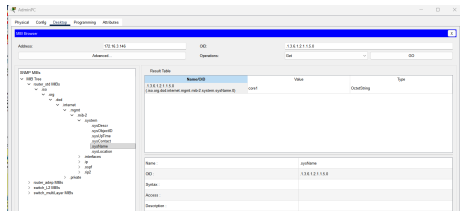


Figure: MIB Browser Result

Project Goals Achieved

- **Scalability:** Hierarchical design supports easy expansion.
- **Redundancy:** Validated OSPF neighbor adjacencies.
- **Connectivity:** Full inter-VLAN communication verified.
- **Management:** Successful SNMP query validation.

Thank You!